

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Calculus I

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) If a function $f(x)$ is differential at a point $x = a$, it is necessarily continuous at that point but the converse may not be true. Prove it. 8

OR

State Leibnitz's theorem. If $y = \sin^{-1} x$, show that;

i. $(1 - x^2)y_2 - xy_1 = 0$

ii. $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$

- b) States Lagrange's mean value theorem and verify it for the function $f(x) = px^2 + qx + r$, $x \in [a, b]$. 7

OR

Apply the Maclaurin's series to find the expansion of $\ln(\sec x)$ as far as the term x^6 and using the series find the expansion of $\tan x$.

2. a) Find the asymptotes of the curves, $x^2y^2 - x^2y - xy^2 + x + y + 1 = 0$. 8
b) Find the radius of curvature at origin of the curves, $y^2 - 2xy - 3x^2 - 4x^3 - x^2y^2 = 0$. 7

3. Attempt any THREE 3×5

a) Evaluate: $\int \frac{dx}{4 + 5\sin^2 x}$

b) Evaluate: $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\cot x}}{1 + \sqrt{\cot x}} dx$

c) What kind of improper integral is $\int_0^2 \frac{1}{(1-x)^2} dx$? Evaluate, if exist.

d) Evaluate by using Gamma function: $\int_0^1 x^4(1-\sqrt{x})^5 dx$

4. a) Find the perimeter of the asteroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$. 7
- b) State and prove Euler's theorem on homogenous functions of two independent variables. If $u = \ln\left(\frac{x^4+y^4}{x+y}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3$. 8
5. a) Determine the maximum and minimum value of $u = 35 - (2x + 3)^2 - (y - 4)^2 - (z + 1)^2$. 7
- b) Define Bernoulli's equation. Solve: $\frac{dy}{dx} - \frac{y}{x} = \frac{y^2}{x^2}$. 8

OR

Suppose that an object is heated to 300°F and allowed to cool in a room whose air temperature is 80°F . After 10 min the temperature of the object is 250°F . What will be its temperature after 20 min?

6. a) Solve: $x^2y'' - 4xy' + 4y = 0$, $y(2) = 4$, $y'(2) = -1$. 7
- b) By the method of variation of parameters solve; $y'' + 2y' + y = 4e^{-x}\ln x$. 8
7. Attempt all the questions. 2.5×4
- a) Examine the continuity of the function $f(x) = \begin{cases} 5-x & \text{if } 1 < x < 2 \\ x^2-1 & \text{if } x \geq 2 \end{cases}$ at $x = 2$.
- b) Evaluate: $\int \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}} dx$
- c) Find the area bounded by the curves $y^2 = 8x$ and $x = 2$.
- d) Find the integrating factor of the equation: $\cos x \frac{dy}{dx} + y \sin x = \sec^2 x$